

E-learning Platform

*A Project Report submitted in partial fulfilment of the
requirements for the award of the degree of*

Master of Computer Applications

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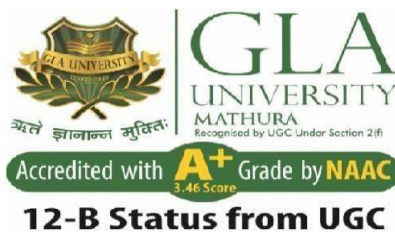
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Under the Guidance of

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GLA University Mathura- 281406, INDIA

April, 2026

Declaration

I hereby declare that the work which is being presented in the M.C.A. Project “**E-Learning Platform**”, in partial fulfillment of the requirements for the award of the **Master of Computer Applications** and submitted to the Department of Computer Engineering and Applications of GLA University, Mathura, is an authentic record of my own work carried under the supervision of **Mr. Mridul Dixit, Assistant Professor**.

The contents of this project report, in full or in parts, have not been submitted to any other Institute or University for the award of any degree.

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ACKNOWLEDGEMENT

We are truly grateful to everyone who guided and helped us throughout this project.

First, we want to thank our project guide, **Mr. Mridul Dixit**, Department of Computer Engineering & Applications, GLA University, Mathura. She guided us at every step — right from planning the idea to completing the final report. Her advice, patience, and encouragement made a big difference in the quality of this project. Whenever we got stuck, she helped us find the right direction.

We are also thankful to the Head of the Department and all our teachers in the MCA programme who taught us the skills we used in this project. Their teaching gave us the knowledge and confidence to take on this work. Their open-door approach whenever we had questions was very helpful.

We thank GLA University for giving us access to good computer labs, the internet, and all the other resources we needed. Having the right tools and a proper working environment made the development process much smoother.

We also want to thank our friends and classmates for their feedback, ideas, and encouragement during the difficult phases of the project. The discussions we had with them helped us think of better solutions and improve our work.

We are thankful to the open-source community — the creators of PHP, MySQL, JavaScript, HTML, CSS, Bootstrap and other tools we used — for making excellent free tools available that helped us build this project.

Finally, we express our deepest thanks to our families for their love, support, and belief in us throughout this journey. Their encouragement kept us motivated even when things got hard.

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ABSTRACT

An e-learning platform is a web-based application designed to provide a digital environment for teaching and learning. The proposed system, Peruse Learning, aims to simplify education by enabling students and instructors to interact through an online interface. This platform allows users to access study materials, video lectures, assignments, and quizzes anytime and from anywhere, making learning more flexible and convenient.

The system is designed with user-friendly features such as course browsing, user registration, login authentication, and personalized dashboards for both students and teachers. Students can enroll in courses, track their progress, and complete assessments, while teachers can upload content, manage courses, and monitor student performance. The platform also supports basic communication between users, enhancing collaboration and engagement.

The frontend of the system is developed using HTML, CSS, and JavaScript to ensure a responsive and interactive user experience across devices such as mobiles, tablets, and desktops. The design focuses on simplicity, accessibility, and effective content presentation. Additionally, dummy data and simulated course content are used to demonstrate the working of the system.

Overall, Peruse Learning aims to improve the traditional learning process by integrating modern technology into education. It provides an efficient, costeffective, and scalable solution that enhances accessibility and promotes self-paced learning. This project demonstrates how digital platforms can transform education and make quality learning resources available to a wider audience.

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List of Abbreviations

BIZ	BizInventory Enterprises
DBMS	Database Management System
DFD	Data Flow Diagram
DML	Data Manipulation Language
DDL	Data Definition Language
DCL	Data Control Language
CLR	Common Language Runtime
SQL	Structured Query Language

Chapter 1 INTRODUCTION

1.1 OVERVIEW AND MOTIVATION (✓)

In today's digital age, education has shifted from traditional classrooms to online platforms because of limitations such as fixed schedules, limited resources, and geographical barriers. Therefore, e-learning systems provide more flexible and accessible learning solutions.

Peruse Learning is a web-based platform that allows students to access courses, study materials, and assessments anytime and anywhere, thus supporting self-paced learning. Moreover, the project is motivated by the growing demand for interactive and user-friendly online education systems.

In addition, it improves communication between teachers and students through features like course management, assignments, and progress tracking. Furthermore, responsive design ensures smooth access across devices.

1.2 OBJECTIVE

The main objective of this project is to design and develop a web-based e-learning platform named *Peruse Learning* that provides an efficient, user-friendly, and accessible environment for online education. The system focuses on improving the learning experience by enabling students and teachers to interact digitally through a centralized platform.

With the rapid growth of online education, one of the major challenges is maintaining student engagement and providing easy access to learning resources. Traditional learning systems often lack flexibility and real-time interaction. This project aims to overcome these challenges by offering a structured platform where students can access courses, study materials, and track their progress, while teachers can manage content and monitor student performance.

The platform is designed to support modern learning needs by providing responsive design, simple navigation, and interactive features. It ensures that users can access the system from any device, making learning more flexible and convenient. Overall, the system aims to enhance digital education by making it more accessible, organized, and effective.

1.3 SUMMARY OF SIMILAR APPLICATION

In recent years, several e-learning platforms have been developed to provide online education and digital learning solutions. These applications offer various features such as video lectures,

course enrollment, progress tracking, and certification. They have significantly transformed the traditional learning system by making education more accessible and flexible.

1.4 ORGANIZATION OF THE PROJECT

The Peruse Learning system is structured to manage online learning activities effectively through different functional components:

Login Module: Ensures secure access using username and password, allowing only authorized students and teachers to use the system.

Course Management: Allows teachers to create and manage courses, upload study materials, and organize content for learners.

Student Dashboard: Enables students to enroll in courses, access materials, and track their learning progress.

Assessment Management: Manages quizzes, assignments, and evaluations to monitor student performance.

Content Management: Provides easy access to video lectures, notes, and other learning resources anytime.

Chapter 2 SOFTWARE REQUIREMENT ANALYSIS

2.1 TECHNICAL FEASIBILITY (✓)

Technical feasibility refers to the analysis of whether the proposed system can be developed using available technology and resources.

For the Peruse Learning system, the development is technically feasible because it is built using widely used technologies such as HTML, CSS, and JavaScript. These technologies are easily available, well-supported, and suitable for developing web-based applications.

The system does not require high-end hardware or specialized tools. It can run on standard computers with basic configurations and can be deployed using local servers if needed. The use of modern web technologies ensures smooth performance, responsive design, and efficient handling of user interactions and content delivery.

2.2 OPERATIONAL FEASIBILITY

Operational feasibility refers to the ability of the system to function effectively in a real-world environment and be accepted by its users.

For the Peruse Learning system, it is operationally feasible because it provides a simple and user-friendly interface that can be easily used by both students and teachers. Users can access courses, study materials, and assessments without any technical difficulty.

The system reduces manual effort by automating tasks such as course management, content delivery, and progress tracking. It also improves communication between students and instructors. Overall, the system is efficient, reliable, and suitable for practical use in the field of online education.

2.3 ECONOMIC FEASIBILITY

Economic feasibility refers to evaluating the cost and benefits of the system.

The Peruse Learning system is economically feasible as it uses free technologies like HTML, CSS, and JavaScript, reducing development cost. It does not require expensive hardware and minimizes costs related to physical classrooms and materials.

Thus, it is a cost-effective solution for online learning

2.4 SYSTEM REQUIREMENTS

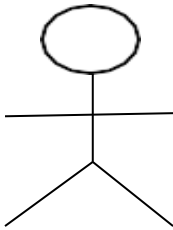
Hardware Requirements:

- Processor: Intel i3 or above□
- RAM: Minimum 4 GB□
- Hard Disk: 100 GB or more□
- Input Devices: Keyboard, Mouse□
- Operating System: Windows □
- Frontend: HTML, CSS, JavaScript□
- Backend: PHP□
- Database: MySQL□
- Server: XAMPP / WAMP□

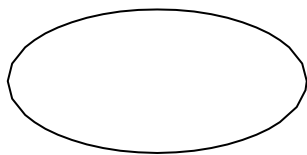
Software Requirements:

2.5 USE CASE

Actor



Use case



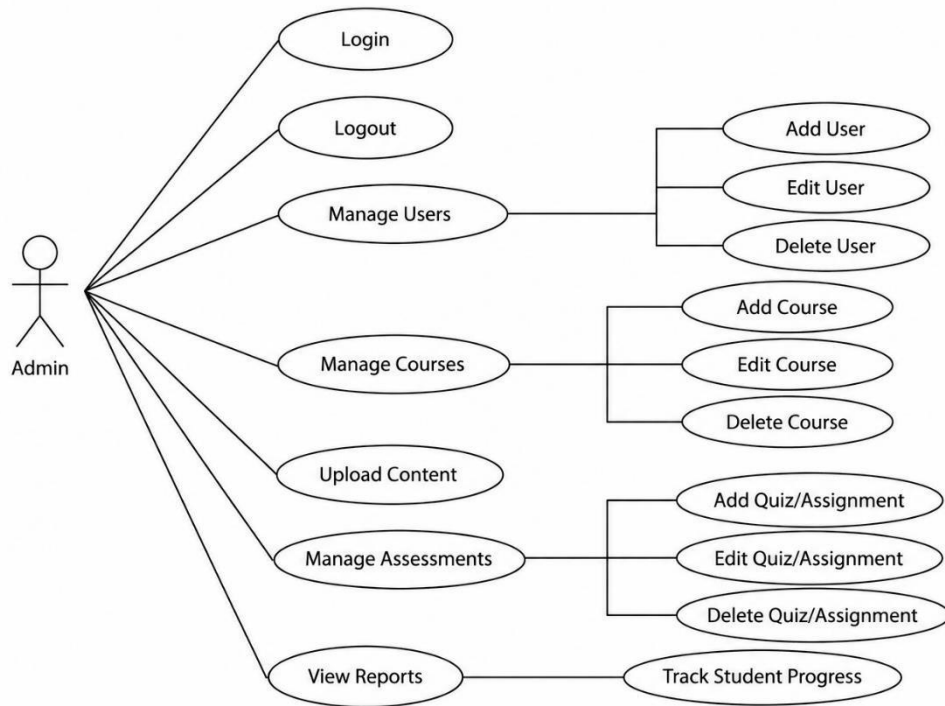


Figure 2.1: E-learning Use Case Diagram

2.6 Functional Requirements

Functional requirements describe what the system should do.

- The system should allow admin login
- The system should manage user details (students and teachers)
- The system should manage courses and study materials
- The system should provide access to video lectures and content
- The system should conduct quizzes and assignments
- The system should track student progress and performance

2.7 Non-Functional Requirements

These define system quality and performance.

- Security: Only authorized users (admin, students, teachers) can access the system
- Performance: The system should provide fast loading and response time
- Usability: Simple and user-friendly interface for easy navigation
- Reliability: The system should provide accurate and consistent data
- Scalability: The system should handle increasing users and data in the future

2.8 System Modules

The system is divided into the following modules:

- Login Module: Handles user authentication (admin, student, teacher)
- User Module: Manages student and teacher details
- Course Module: Manages courses and study materials
- Content Module: Provides video lectures and learning resources
- Assessment Module: Manages quizzes and assignments
- Progress Module: Tracks student performance and progress

Chapter 3 SOFTWARE DESIGN

3.1 SYSTEM DESIGN OVERVIEW

The Peruse Learning system is designed as a web-based application that manages online learning activities efficiently. The system follows a modular structure where different components handle specific tasks such as login, course management, content delivery, and assessment handling.

The system is designed to be secure, efficient, and easy to use. It supports learning operations like accessing courses, viewing study materials, and completing assessments with real-time updates. The modular design also makes the system easy to maintain and upgrade in the future.

3.1.1 DATA FLOW DIAGRAM (DFD)

The software design of the Peruse Learning system defines the overall architecture and structure of the system. It explains how data flows within the system, how different components interact, and how the content and user data are managed.

The system is designed in a modular way, where each module performs a specific function such as login, course management, content delivery, and assessment handling. The design ensures scalability, security, and easy maintenance.

Level 0 DFD (Context Diagram)

This is the top-level diagram showing the interaction between the system and external entities.

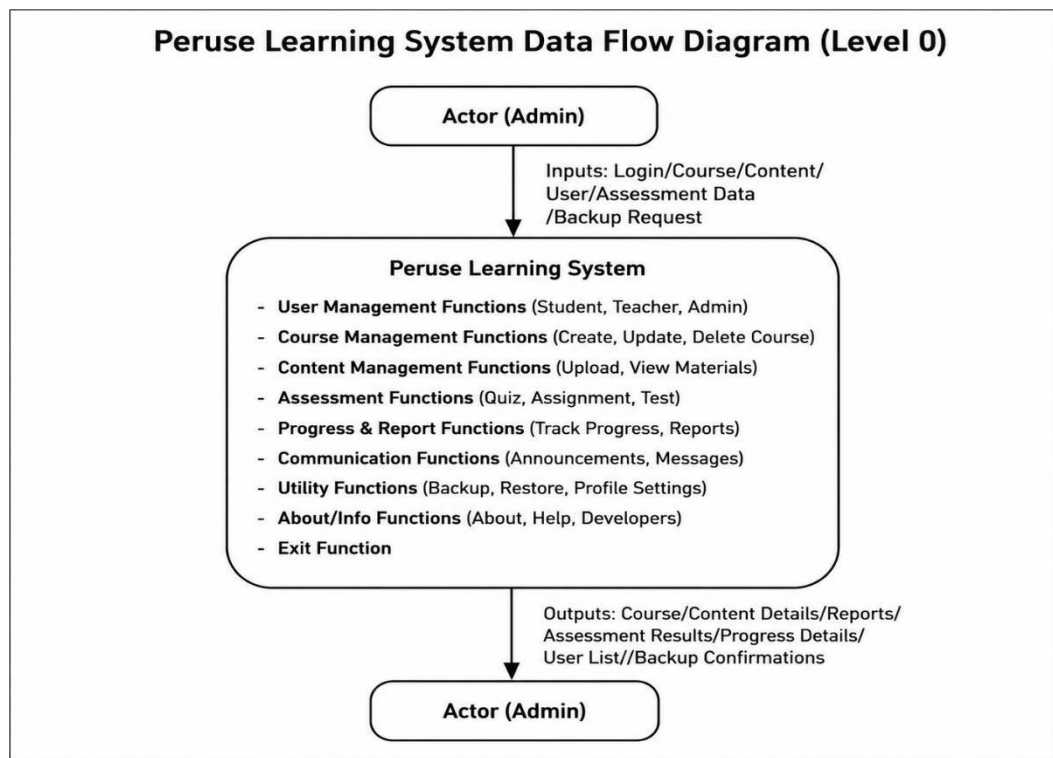


Figure 3.1.1: Level 0 DFD

Level 1 DFD

This diagram breaks the system into smaller processes.

The Level 1 DFD divides the system into multiple processes such as Login Management, Stock Management, Sales Management, and Purchase Management. Each process interacts with the database to store and retrieve information.

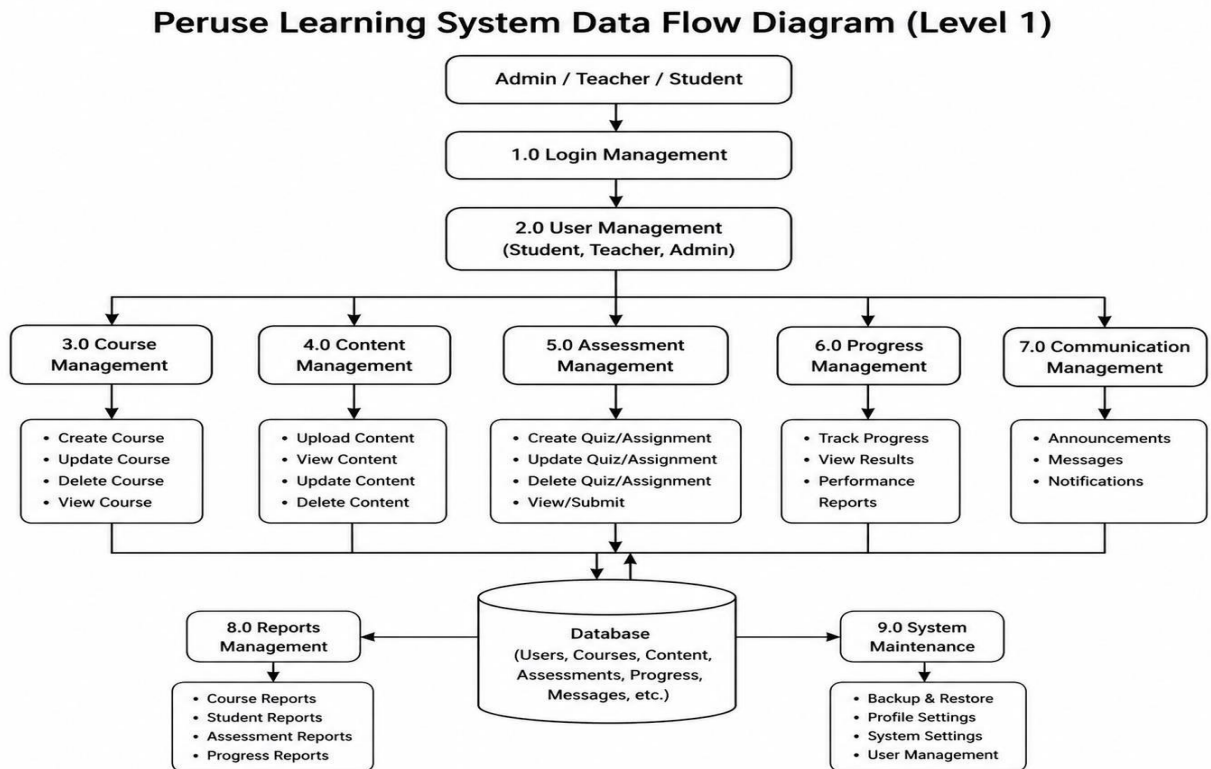


Figure 3.1.2: Level 1 DFD

3.1.2 UML DIAGRAMS

The **Unified Modeling Language (UML)** diagrams are used to visualize, specify, and document the design of the **E-learning Platform**. These diagrams help in understanding the structure and behavior of the system by representing different components and their interactions.

UML diagrams play an important role in software development as they provide a clear blueprint of the system before implementation. In this project, various UML diagrams such as Class Diagram, Object Diagram, Sequence Diagram, and Collaboration Diagram are used.

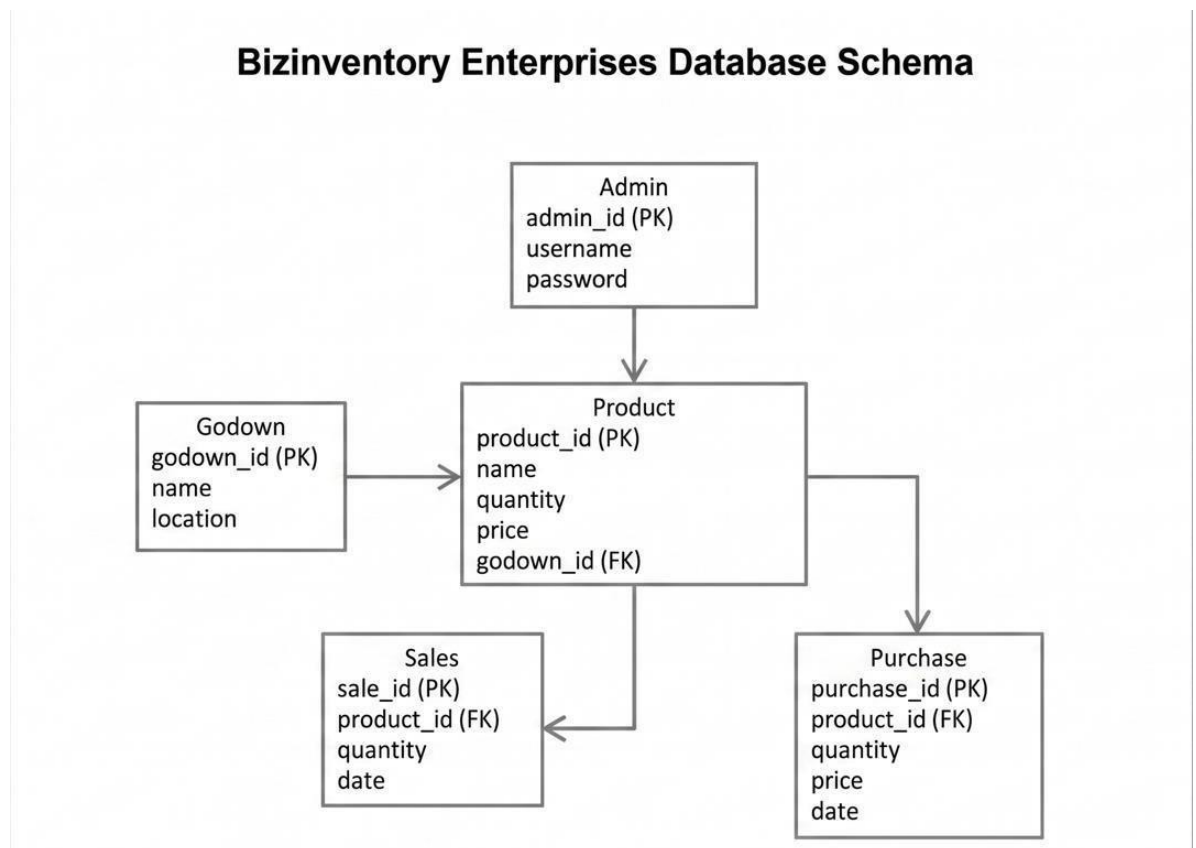


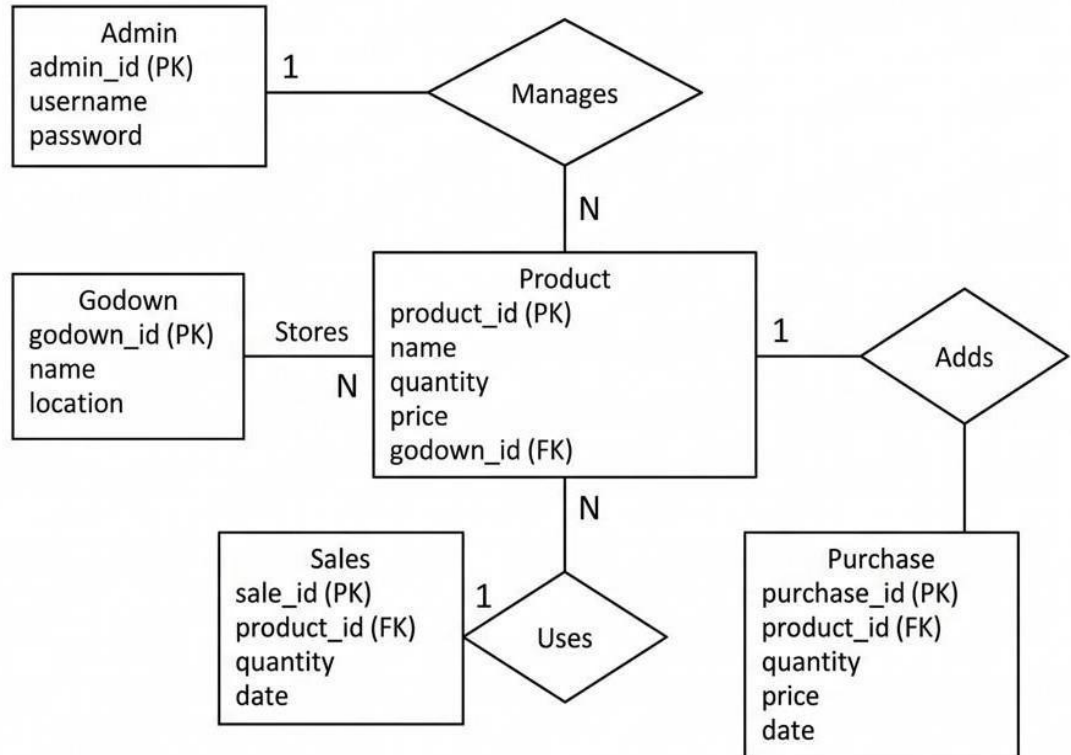
Figure 3.2: UML Diagram

3.1.3 DATABASE DESIGN

The database design of the **BizInventory Enterprises** system defines how data is organized, stored, and managed efficiently. A well-structured database ensures data consistency, accuracy, and easy retrieval of information.

The system uses a **relational database model** implemented using MySQL. In this model, data is stored in the form of tables, and relationships are established between them using keys.

BizInventory Enterprises - Sales, Stock, and Vendor Schema



3.1.3 DATABASE DESIGN

3.1.4 FLOW CHART DIAGRAM

The flowchart diagram represents the step-by-step flow of the system processes in a graphical form. It helps in understanding how the system works logically from start to end.

In the E-learning, the flowchart describes the sequence of operations such as user login, managing stock, recording sales and purchases, and generating reports.

Flowcharts use standard symbols like:

- Oval → Start/End
- Rectangle → Process
- Diamond → Decision
- Arrow → Flow of control

The flowchart provides a clear view of how data moves through the system and how different operations are performed.

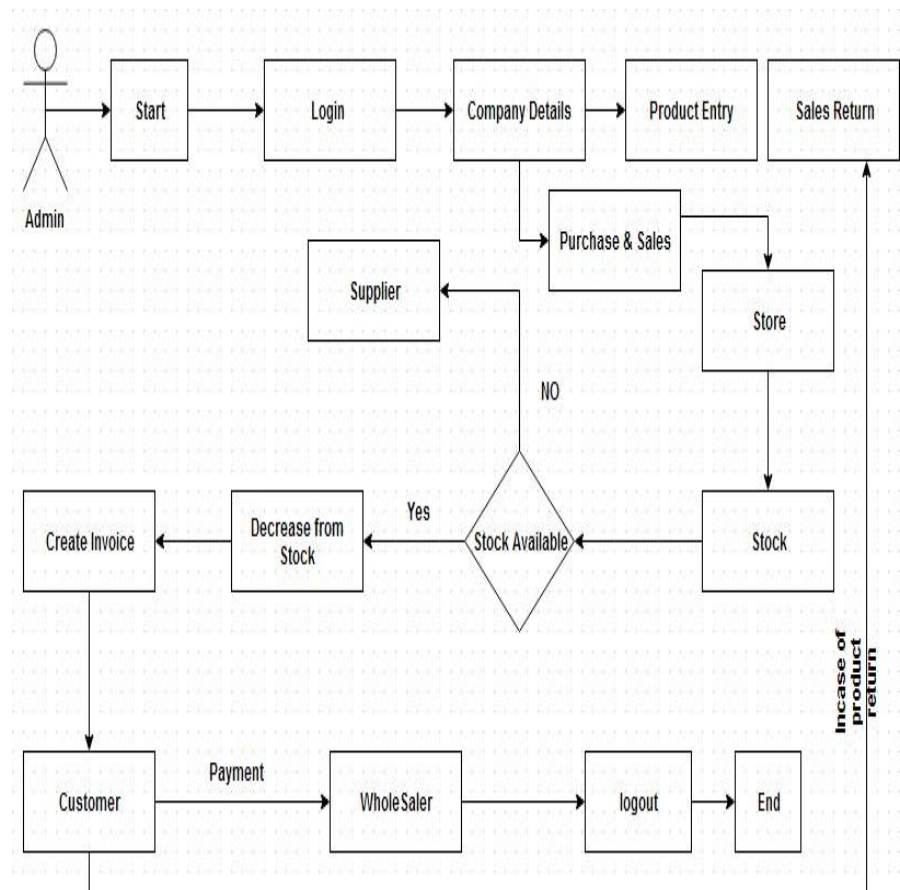


Figure 3.1.4 : BIZ Process flow diagram

Chapter 4 IMPLEMENTATION AND USER INTERFACE

The **E-learning Platform** is implemented as a web-based application using frontend technologies such as **HTML, CSS, and JavaScript**. The system is designed to provide an interactive and user-friendly environment for accessing online courses and learning resources.

The frontend of the system is developed using HTML for structure, CSS for styling, and JavaScript for adding interactivity. Different pages such as the home page, course page, login/signup page, and dashboard are created to manage user navigation and provide a smooth experience.

The platform currently works using static content and simulated data, where course details and user information are displayed through predefined data

Peruse learning

. This helps demonstrate how a real e-learning system functions even without backend integration.

The user interface is designed to be simple and easy to use. It includes clearly structured sections such as header, body, and footer, along with proper navigation menus. The design is also responsive, allowing the application to run smoothly on mobile, tablet, and desktop devices.

The system has been tested under different conditions to ensure proper working of features such as navigation, form validation, and user interaction. The modular structure of the project makes it easy to maintain and enhance in the future.

4.1 ADMIN OR LOGIN PAGE

The **Login Page** acts as the entry point of the **E-learning Platform**. It allows users (students or admin) to access the system by entering a valid username and password.

This page plays an important role in maintaining security by restricting unauthorized access. If the entered credentials are incorrect, the system displays an error message. If the credentials are correct, the user is successfully redirected to the dashboard.

The login page is designed using simple and user-friendly elements such as input fields, buttons, and validation messages. Basic form validation is implemented using JavaScript to ensure that users enter the required information before submitting the form.

Currently, the system uses dummy user data for login functionality

Peruse learning

, but it can be enhanced in the future by integrating a database for secure authentication and user management.

Overall, the login page ensures secure access and provides a smooth starting point for users to interact with the platform.

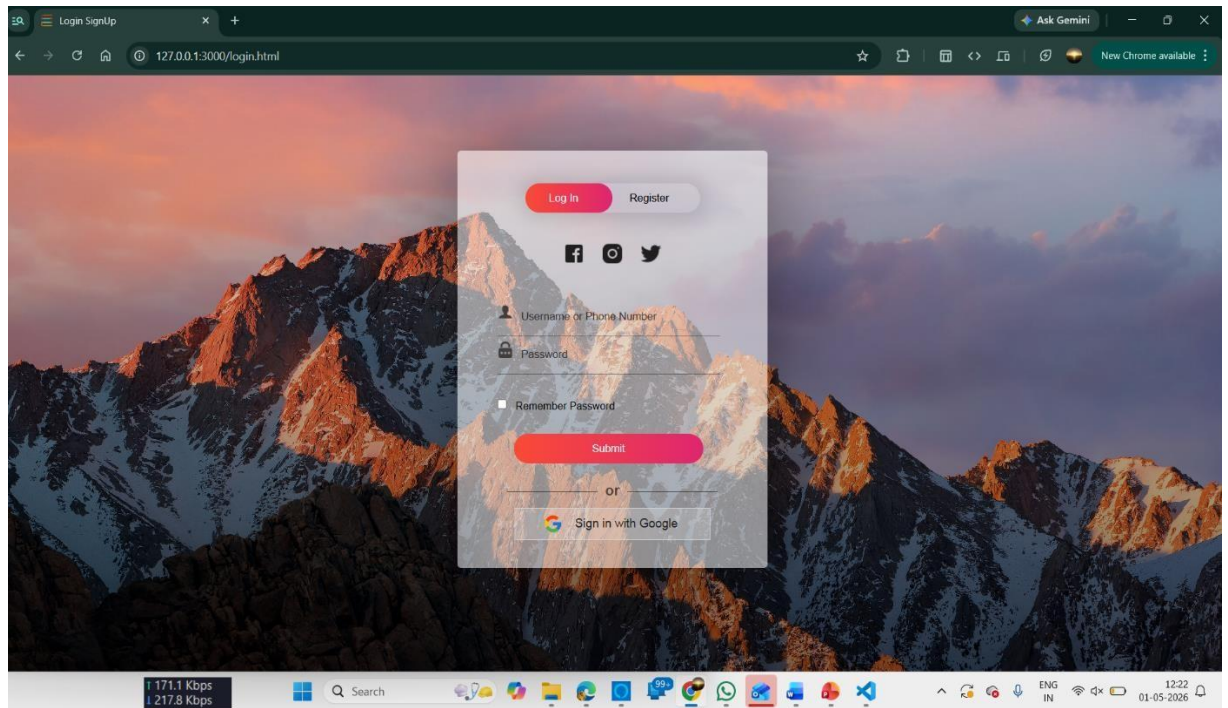


Figure 4.1: Admin or login page

4.2 Home Page Panel

The **Home Page Panel** is the main interface of the **E-learning Platform**. It serves as the central point from where users can access different features of the system.

This panel provides an overview of the platform, including available courses, featured content, and important sections such as login, signup, and navigation menus. It allows users to quickly move to different pages like courses, dashboard, and contact page.

The Home Page is designed to be simple, interactive, and user-friendly. It includes elements such as banners, course cards, and navigation links to improve the overall user experience. The layout is structured with header, body, and footer sections for better organization.

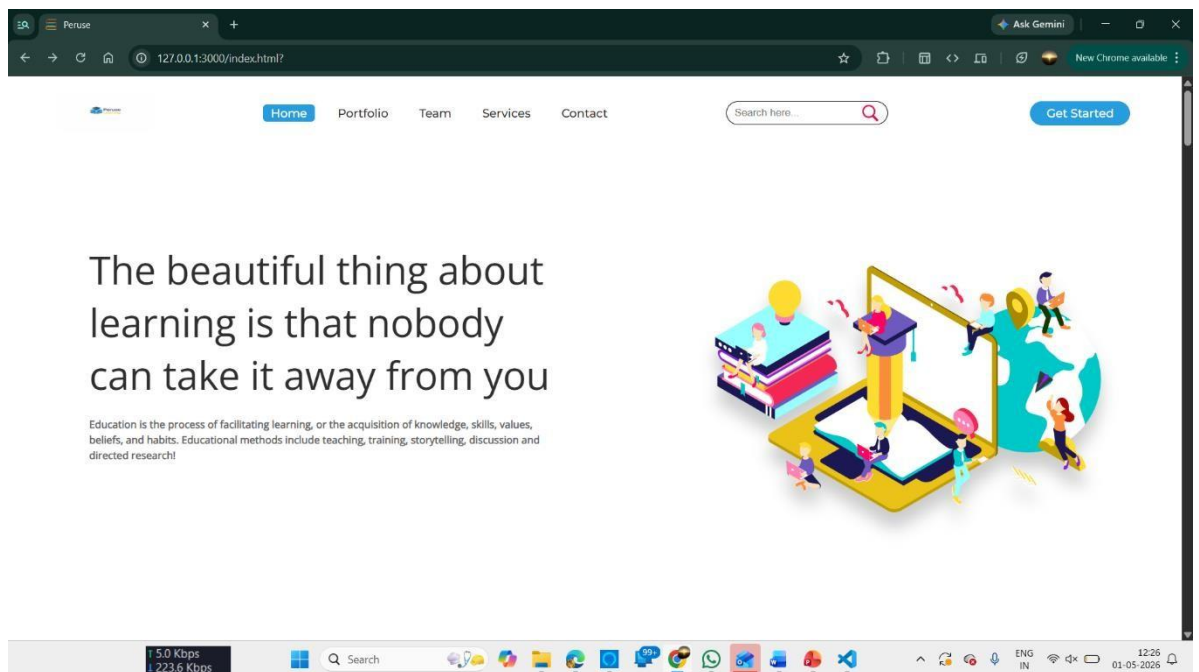


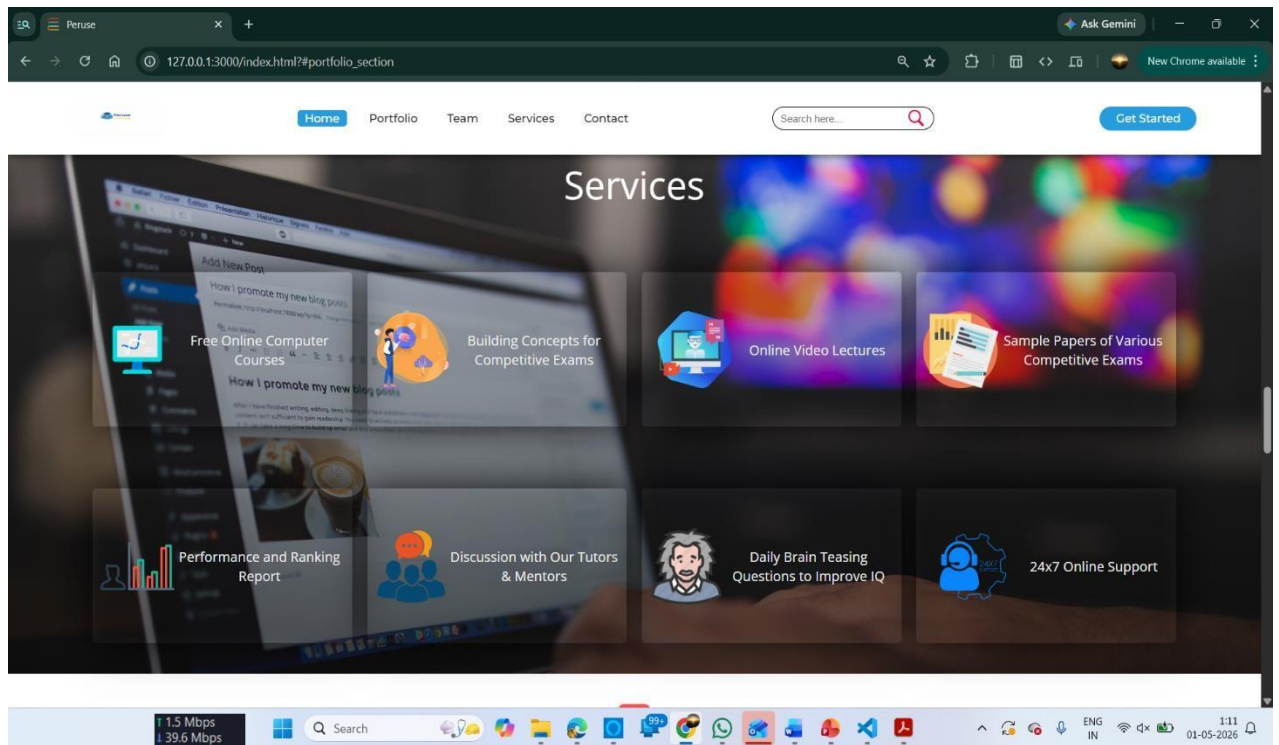
Figure 4.2: Home Page

4.3 Services Panel

The **Services Panel** is used to manage and display different services or course categories in the **Elearning Platform**. It allows the administrator to add new services (courses), update existing details, and remove services when required.

This panel helps in organizing courses based on different categories such as programming, design, or business. It makes the platform more structured and improves the user's ability to search and explore courses easily.

The Services Panel also enhances content management by allowing clear classification of courses, which improves navigation and overall user experience. It ensures that users can quickly find relevant 1



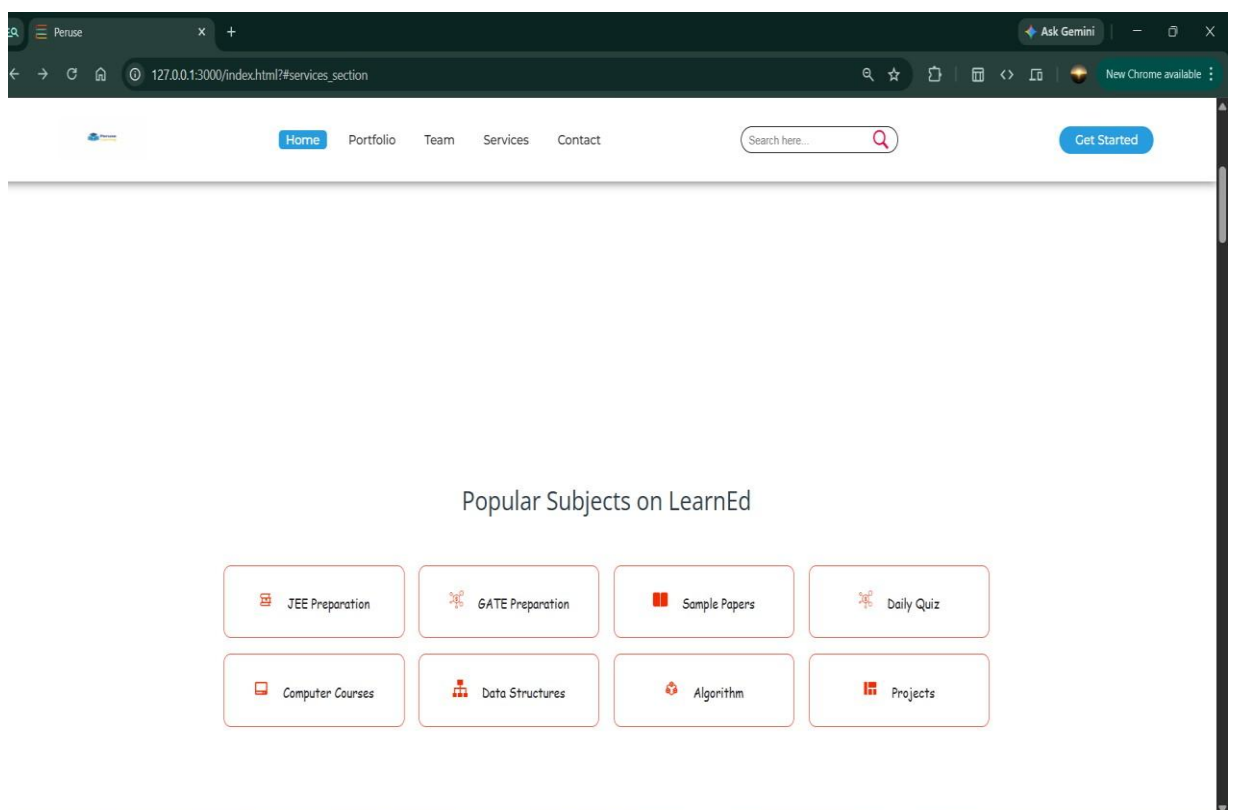
4.4: CATEGORIES PANALS

The **Categories Panel** is used to manage different course categories in the **E-learning Platform**. It allows the administrator to add new categories, update existing ones, and delete categories when required.

This panel helps in organizing courses into different groups such as programming, design, management, and other subjects. It makes the system more structured and improves the overall organization of learning content.

The Categories Panel also enhances searching and filtering, allowing users to quickly find courses based on their interests. Proper classification of courses improves navigation and makes the platform more user-friendly.

Overall, this panel plays an important role in maintaining an organized system and improving accessibility and usability of the e-learning platform.



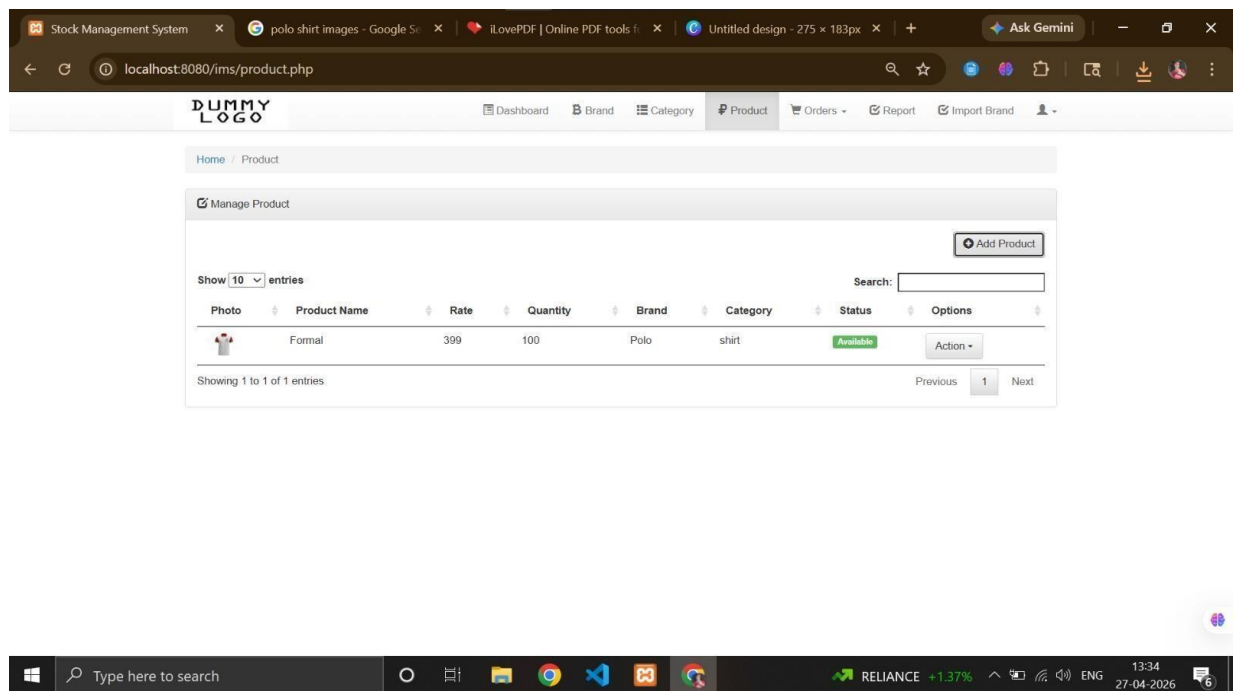


Figure 4.5: Product Panels

4.5 : ORDER PANALS

The Order Panel is used to manage all customer orders in the BizInventory Enterprises system. It allows the administrator to create new orders, view existing orders, and update order details as required.

This panel records important information such as product details, quantity ordered, total amount, and order date. It helps in tracking sales transactions and ensures proper management of order records.

The Order Panel also plays a crucial role in updating stock levels automatically when an order is placed, ensuring accurate inventory management.

4.6: GENERATE REPORT PANAL

The Report Panel in the Inventory Management System is an important module used to generate, manage, and analyze various inventory-related reports. It provides users with detailed information about stock levels, product availability, sales records, purchase history, and inventory movement. Through this panel, users can monitor low stock items, track incoming and outgoing products, and review supplier or transaction details. It helps maintain accurate records and supports better control over inventory operations. The report panel also assists in identifying trends, reducing errors, preventing stock shortages or overstocking, and improving overall efficiency. By presenting organized and meaningful data, it supports informed decision-making and helps managers evaluate the performance of the inventory system effectively.

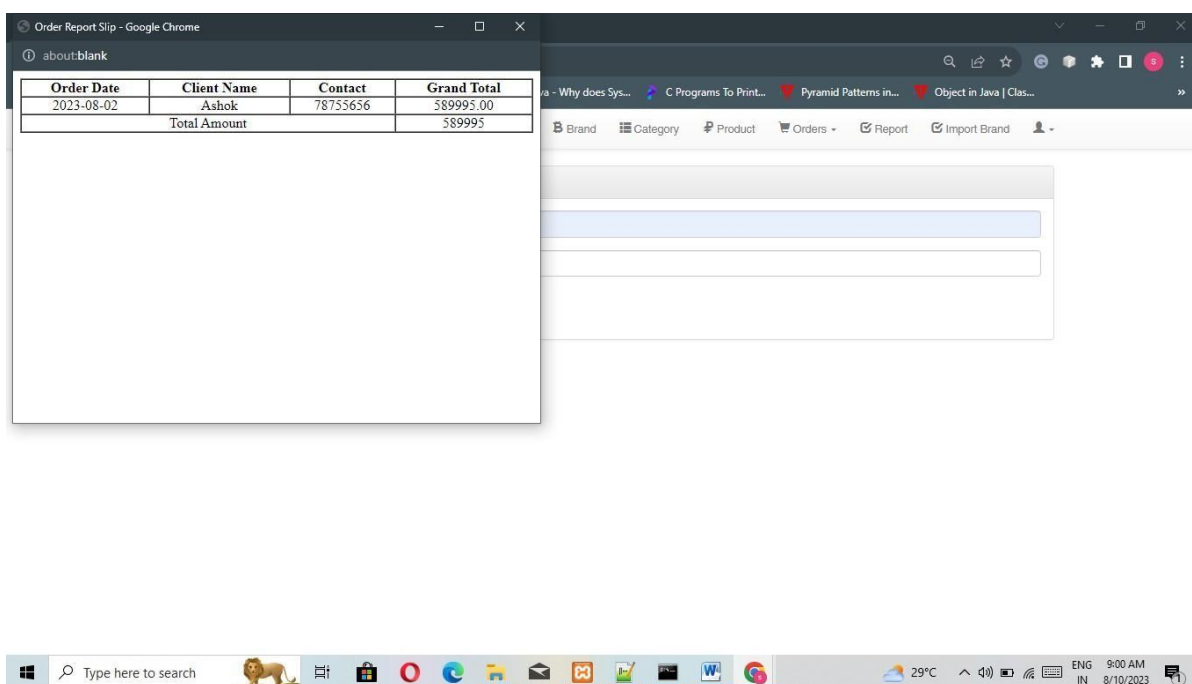


Figure 4.8: Generate Report Panal

CHAPTER 5

TESTING

5.1 : PURPOSE OF TESTING

The purpose of software testing is to evaluate the performance and functionality of a software application and to ensure that it meets the required standards and user needs. Testing helps in identifying defects and verifying that the system works correctly under different conditions.

Testing involves both **verification and validation**, which ensure that the application is developed correctly and fulfills its intended purpose. The main objective of testing is to detect errors, improve reliability, and provide useful information about the system's performance.

In this project, testing is performed to ensure the proper functioning of the application. It checks whether all operations such as inserting, updating, and deleting data are working correctly. It also ensures that the user interface behaves as expected and provides a smooth experience.

Overall, testing plays an important role in improving the quality, reliability, and usability of the application.

5.1.1 :TYPE OF TESTING

To ensure that the application works correctly and is free from major errors, we have applied **unit testing** as the primary testing method. This type of testing helps verify the functionality of individual components of the system.

5.1.2 :UNITS TEST

Unit testing involves testing individual modules or components of the software separately. It is usually performed by the developer, as it requires a good understanding of the program's internal structure and code.

During unit testing, we tested different parts of the application such as form inputs, buttons, and page functionalities. We verified whether data is correctly processed and reflected in the system interface. Various bugs related to individual components were identified and fixed during this phase.

We also tested different functional operations to ensure that features like navigation, data handling, and user interactions are working properly. This helped improve the reliability and performance of the application.

CHAPTER – 6

CONCLUSION

6.1 : PROJECT LIMITATION

Since this is our first project, it has some limitations due to limited time and experience. Some expectations could not be fully achieved during development. However, these limitations are understandable and can be improved in future versions.

The main limitations of the project are:

- The application is not suitable for large-scale organizations with complex requirements.
- It supports only basic features and simple functionality.
- The system uses static or dummy data instead of a real database.
- Only a single user/admin panel is available.
- It does not provide advanced reports or analytics.

6.2 : CONCLUSION

To conclude, the **E-learning Platform** is a simple web-based application suitable for basic educational needs. It includes essential features such as course pages, login/signup, and dashboards. The system allows users to access learning content easily and interact with the platform in a structured way.

Our team successfully developed a platform where users can explore courses and perform basic operations. The application provides a smooth user interface and demonstrates how online learning systems work.

Although the project has some limitations, such as lack of backend and real-time features, it still serves as a strong foundation. We believe that with future improvements, this system can greatly benefit students and educational organizations.

6.3 : LESSON LEARNT

Doing a project over a long period always provides valuable learning experiences. During the development of this project, our team learned several important concepts and skills.

Some of the key lessons learnt are:

- Gained knowledge of frontend technologies such as HTML, CSS, and JavaScript
- Learned how to design a user-friendly and responsive interface
- Understood the importance of proper planning and project structure
- Improved problem-solving and debugging skills
- Developed teamwork and communication skills
- Learned how to manage time effectively during project development

CHAPTER 7

SUMMARY

In addition to its core functionalities, the **E-learning platform** focuses on organized content management and structured presentation of educational resources. All course-related information such as course name, description, instructor details, and content modules are arranged systematically within the system. This structured approach helps users easily browse and access courses while reducing confusion and improving overall usability.

Another important feature of the platform is its interactive and responsive user interface. The system is designed using HTML, CSS, and JavaScript, ensuring smooth navigation across different pages like home, courses, login/signup, and dashboard. The responsive design allows users to access the platform on mobile, tablet, and desktop devices without any difficulty. This improves accessibility and provides a better learning experience.

The project also emphasizes simplicity and ease of use. Each module is clearly organized so that users can quickly understand how to navigate through the platform. Features like course listing, user login, and dashboards are implemented in a straightforward manner, making the system suitable for beginners. The use of basic JavaScript functions such as button actions and form validation enhances interactivity and user engagement.

From a functionality perspective, the system simulates real-world operations using dummy data. Course cards, user details, and static content are used to demonstrate how an actual e-learning system would work. This helps in understanding the workflow of online learning platforms even without backend integration.

The platform is flexible and can be adapted for different types of educational use cases such as schools, coaching institutes, and self-learning portals. Being a web-based system, it can be accessed easily through a browser without requiring complex installations, making it convenient for users.

Looking ahead, the project has strong potential for future enhancements. Features such as database integration, real-time user authentication, video lectures, live classes, and AI-based course recommendations can be added. Additional improvements like performance tracking, online assessments, and payment integration can further enhance the system's capabilities.

Overall, the Peruse Learning project successfully demonstrates how web technologies can be used to build a basic e-learning platform. It highlights the importance of accessibility, user-friendly design, and digital learning solutions in modern education. The project serves as a strong foundation for developing a complete and advanced Learning Management System (LMS) in the future.

APPENDICES

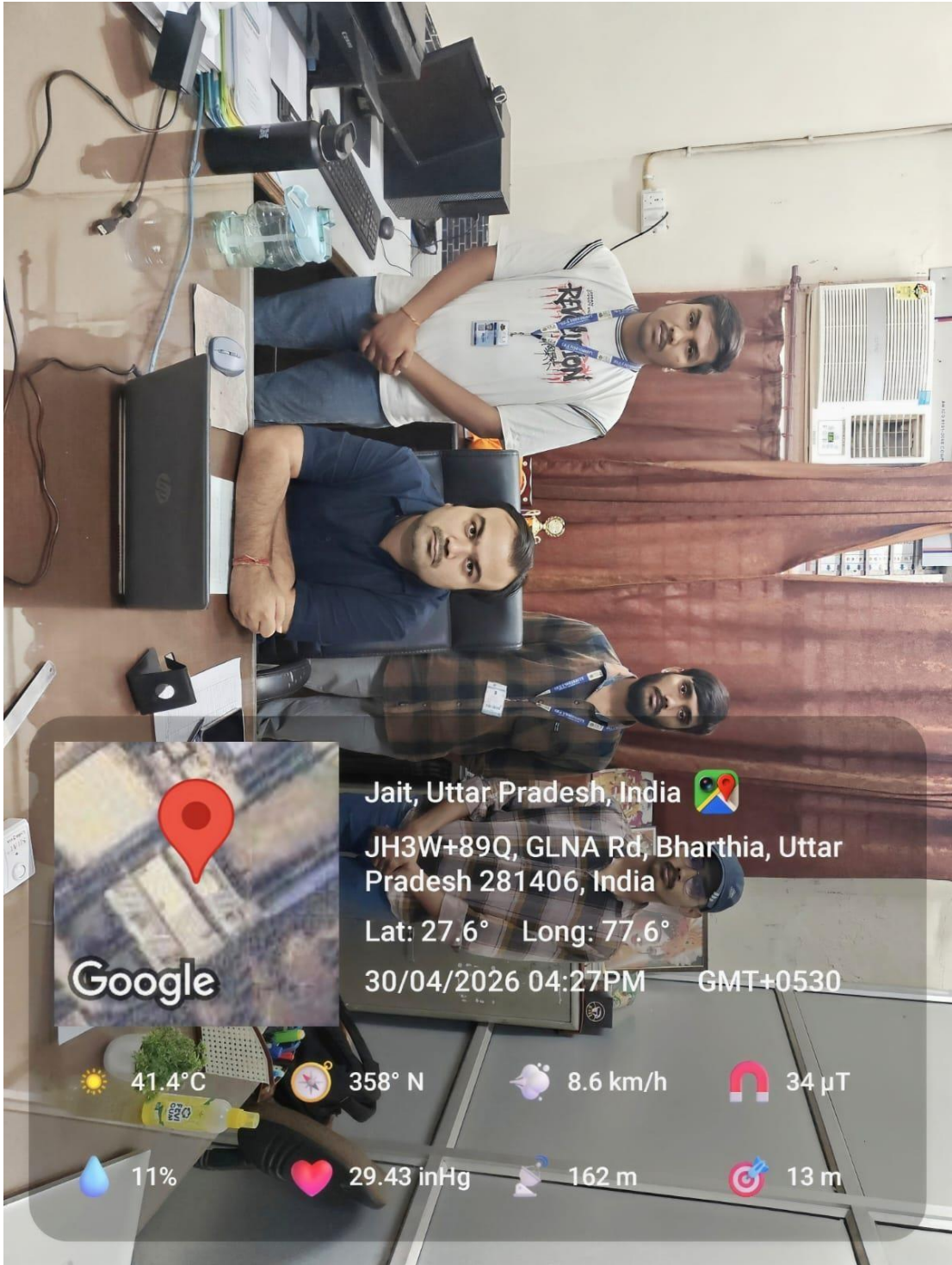
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APPENDIX E: LIMITATIONS OF THE SYSTEM

The current system has several limitations that can be addressed in future enhancements:

- The system is limited to single-user access in its basic version, restricting collaborative usage.
- There is no cloud integration, which limits remote accessibility and scalability.
- Graphical reporting features are limited, reducing advanced data visualization capabilities.
- The system does not support mobile applications, making it less accessible on smartphones and tablets.
- Security implementation is basic and lacks advanced protection mechanisms such as encryption and multi-factor authentication.



Jait, Uttar Pradesh, India 📍

JH3W+89Q, GLNA Rd, Bharthia, Uttar Pradesh 281406, India

Lat: 27.6° Long: 77.6°

30/04/2026 04:27PM GMT+0530



41.4°C



358° N



8.6 km/h



34 μ T



11%



29.43 inHg



162 m



13 m

